

Minglai Yang

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Education

University of Arizona

B.S. in Computer Science

Tucson

2023–2025

- GPA: 4.0 / 4.0
- Expected Graduation: Fall 2025 (Accelerated completion of 4-year curriculum in 2 years)
- All relevant coursework available [here](#).

Research Experience

University of Arizona, CLU-LAB

Undergraduate Research Assistant

Tucson, AZ

Oct 2024 – Present

- Advisor: Prof. Mihai Surdeanu, Dr. Liangming Pan and Dr. Steven Bethard
- Led Research Project [1] on analyzing LLM reasoning, designed GSM-DC, a controlled benchmark for evaluating language models reasoning under irrelevant context, revealing **6 new empirical laws** on how irrelevant context disrupts reasoning, which training regimes improve robustness, and proposing test-time methods that achieve up to **6.3%** out-of-distribution performance gains (**EMNLP Oral – First Author**).
- Conducting research with Dr. Vikas Yadav on speculative decoding strategies for LLMs, introducing CopySpec [2], a copy-and-verify approach that delivers up to **3.08× faster generation** and **2.66× higher efficiency** in LLM self-correction reasoning (**EMNLP Oral – Second Author**).
- Leading a project with Prof. Steven Bethard on long-context LLM reasoning, investigating whether model reasoning is trustworthy versus prone to hallucination and other failure modes (**SemEval In Submission – First Author**).
- Collaborated with Dr. Chicheng Zhang to develop LLM-guided warm starts for RL [3], achieving at least 20× higher data efficiency (**EXAIT@ICML Workshop – Second Author**).
- Proposed using PPO to optimize the HIT@DICT reward for relation extraction; our RL optimization raised one-shot NYT29 F1 **from 24.65% to 48.11%**, with more stable convergence (**ICLR In Submission – Third Author**).

University of Arizona, ML4AI-LAB and IVI-LAB

Undergraduate Research Assistant

Tucson, AZ

Oct 2023 – Present

- Advisor: Prof. Kobus Barnard and Dr. Adarsh Pyarelal
- Developing dynamic Bayesian networks and Theory of Mind models within the **ToMCAT** project to improve AI modeling of human coordination.
- Leading a project on evaluating LLM planning capabilities under adversarial conditions, designing a controlled testbed to conduct stepwise analysis and benchmarking.
- Designing probabilistic models to disentangle causal interpersonal coordination from correlation.
- Co-authored a \$544K, two-year research proposal, focusing on LLM planning, temporal-causal reasoning, and robustness evaluation against noisy commands.

Tsinghua University, Knowledge Engineering Group

Visiting Student

Beijing, China

May 2025 – Aug 2025

- Advisor: Prof. Juanzi Li
- Directed a research project on mechanistic interpretability of reasoning processes in reinforcement learning-based LLMs.
- Collaborating on a survey paper on the mechanisms of multi-step reasoning in LLMs (**TACL In Submission**).

Publications

1. *How Is LLM Reasoning Distracted by Irrelevant Context? An Analysis Using a Controlled Benchmark*
Minglai Yang, Ethan Huang, Liang Zhang, Mihai Surdeanu, William Wang, Liangming Pan
EMNLP 2025 — Main Conference [**Oral**] (CORE A*), 2025 [[preprint](#)] [[code](#)] [[slides](#)] [[poster](#)] [[project page](#)]
2. *CopySpec: Accelerating LLMs with Speculative Copy-and-Paste Without Compromising Quality*
Razvan Dumitru, **Minglai Yang**, Vikas Yadav, Mihai Surdeanu
EMNLP 2025 — Main Conference [**Oral**] (CORE A*), 2025 [[preprint](#)] [[code](#)]
3. *Improving the Data-efficiency of Reinforcement Learning by Warm-starting with LLM*
Thang Duong, **Minglai Yang**, Chicheng Zhang
Exploration in AI Today (EXAIT) Workshop @ ICML, 2025 [[preprint](#)] [[code](#)]

Preprints

1. *AlignSAE: Concept-Aligned Sparse Autoencoders*
Minglai Yang, Xinyu Guo, Mihai Surdeanu, Liangming Pan
ACL In Submission, 2026 [\[preprint\]](#)
2. *Answer or Abstain: Evidence-Calibrated RAG*
Minglai Yang, Steven Bethard
SemEval In Submission, 2026
3. *Peeking inside the Black-Box: Reinforcement Learning for Explainable and Accurate Relation Extraction*
Xinyu Guo, Zhengliang Shi, **Minglai Yang**, Mihai Surdeanu
ICLR In Submission, 2026 [\[preprint\]](#)[\[code\]](#)

Working Experience

Coretechs

Kensington, MD

Machine Learning Engineer Intern (Full Time)

May 2024 – Aug 2024

- Developed GPT-powered Slack bots with retrieval-augmented generation (RAG) and integrated real-time Slack chat into the official website using JavaScript, increasing active user engagement by 57%.
- Led end-to-end website development, from design to deployment, leveraging AWS (EC2, RDS, S3), Nginx, and Docker for scalable infrastructure.
- Conducted performance testing with Selenium, simulating 500+ users to ensure system reliability under high load.
- Received the *Best Intern Award* for outstanding contributions to the company's first GenAI project.

The University of Arizona, AI Core

Tucson, AZ

Lead AI Engineer (Part Time)

Feb 2024 – Dec 2024

- Built tool-calling AI agents with Retrieval-Augmented Generation (RAG) for internal knowledge; trained a custom long-context documentation QA model, improving top-1 accuracy by 48% on internal benchmarks.
- Designed and shipped an end-to-end campus chatbot website serving 50,000+ UA students with long-context retrieval; increased student web engagement by $5.39\times$ post-launch.

Honors and Awards

2025: 2nd Place — **Reddit Wildcat Hackathon** *Reddit & University of Arizona*

2025: **Undergraduate Research Travel Grant (Top 10 awardees university-wide)**, *University of Arizona*

2025: **Galileo Circle Scholar (Top 0.8% Academic Award)**, College of Science, *University of Arizona*

2023–2025: **Academic Highest Distinction (Dean's List) (6 times)**, *University of Arizona*

2023–2025: **Global Wildcat Scholarship**, *University of Arizona*

2021: **Shanghai High School Science and Technology Competition**, Second Prize, *Shanghai High School*

2020: **Shanghai Youth Science and Technology Innovation Competition**, Third Prize, *Shanghai, China*

Leadership and Extracurricular Activities

AI Club, University of Arizona

Tucson, AZ

President

Aug 2024 – Present

- Advisor: Dr. Zhuolin Yang, Dr. Adarsh Pyarelal and Prof. Kobus Barnard
- Organized workshops on AI agents and LLM applications, including a Hack Arizona event with 100+ participants.
- Led weekly lectures and invited speaker sessions, engaging over 40 students across disciplines to foster AI learning.
- Organized a reading group that mentors students interested in AI research.
- Raised over \$12,000 in sponsorships to support club activities and student research projects.

Mathematical Contest in Modeling (MCM), University of Arizona

Tucson, AZ

Team Leader & First Author

Sep 2023 – Mar 2024

- Led weekly tutoring sessions on mathematical modeling, convex optimization, Bayesian inference, and deep neural network methods.
- Developed mathematical models for the 2024 MCM problem on sea lamprey ecosystems, focusing on resource availability and sex ratio dynamics.

Student Union at University of Arizona

Tucson, AZ

Student Media Worker

Aug 2023 – Feb 2024

- Hosted large-scale showcase and orientation events, welcoming over 1,000 new students to the university.
- Organized and supported student engagement programs, strengthening outreach and inclusion on campus.

Teaching

Fall 2025: TA, CSC-244: Discrete Mathematics II, Computer Science, University of Arizona

Fall 2025: Instructor, Deep Learning and Natural Language Processing, AI Club, University of Arizona

Spring 2025: TA, CSC-144: Discrete Mathematics I, Computer Science, University of Arizona

Fall 2024 – Spring 2025: Instructor, Math for AI Workshop Series, AI Club, University of Arizona

Service

AI4MATH Workshop @ ICML

Reviewer - RL Post-training

May 2024 – Jun 2024

Technical Skills

Languages & Systems: Python, Java, C/C++, JavaScript, MATLAB, Ruby, Swift; Linux, Git, Docker, AWS, Nginx

ML/DL & Finetuning: PyTorch, NumPy, pandas, scikit-learn, PyMC; PEFT (LoRA/QLoRA, Prefix/P-Tuning), SFT/DPO/PPO, RLHF/RLAIF; LLaMA-Factory, VERL, OpenRLHF, DeepSpeed-Chat; bitsandbytes

NLP & RAG/Serving: Hugging Face (Transformers/Datasets/Tokenizers), spaCy, NLTK, SentencePiece, SentenceTransformers; LangChain, LlamaIndex, FAISS, Milvus, Qdrant, Weaviate, Chroma, pgvector; Elasticsearch/BM25, vLLM, FlashAttention, TransformerLens